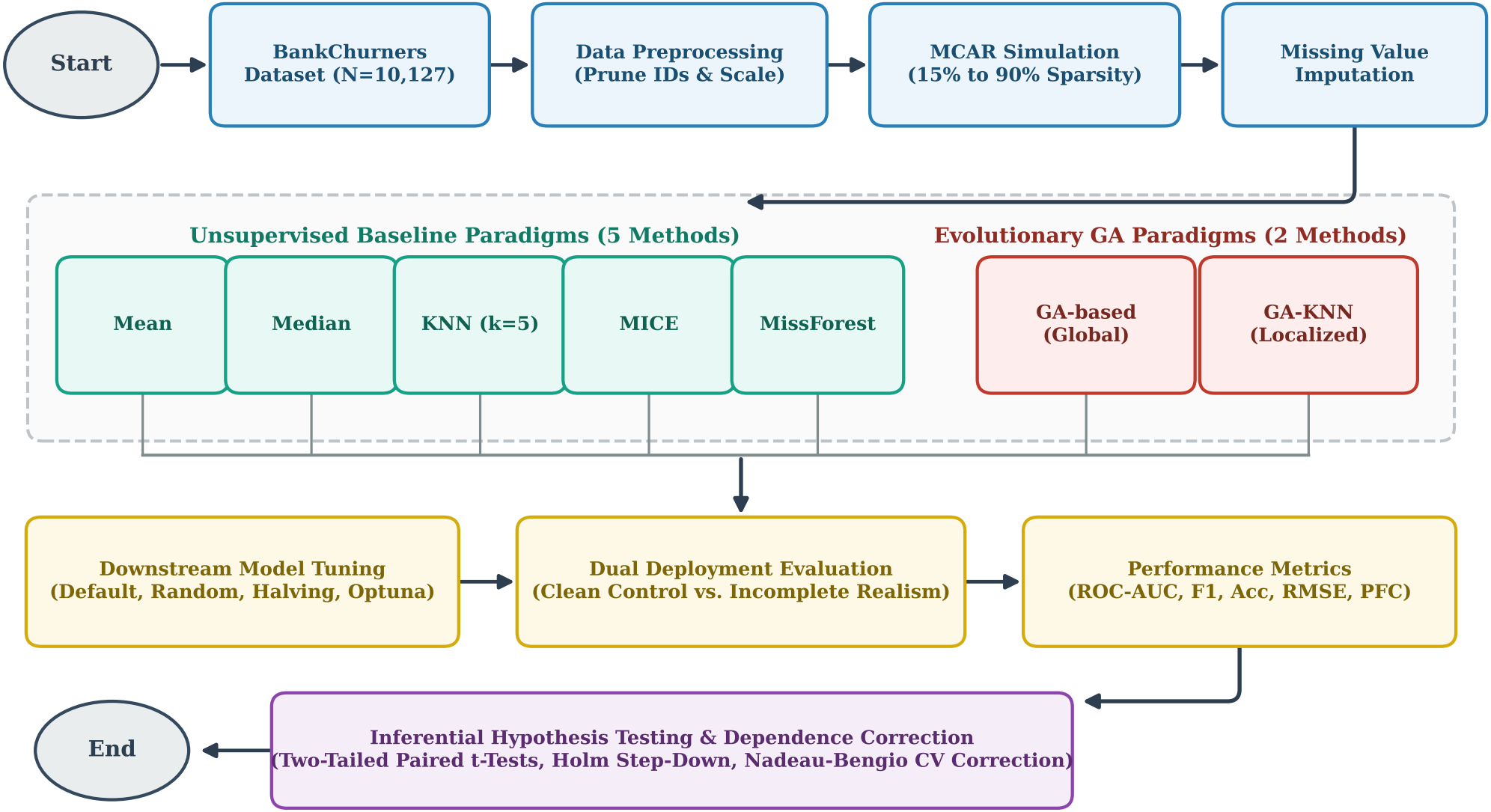
Start
BankChurners Dataset (N=10,127)
Data Preprocessing (Prune IDs & Scale)
MCAR Simulation (15% to 90% Sparsity)
Missing Value Imputation
Unsupervised Baseline Paradigms (5 Methods)
Mean
Median
KNN (k=5)
MICE
MissForest
Evolutionary GA Paradigms (2 Methods)
GA-based (Global)
GA-KNN (Localized)
Downstream Model Tuning (Default, Random, Halving, Optuna)
Dual Deployment Evaluation (Clean Control vs. Incomplete Realism)
Performance Metrics (ROC-AUC, F1, Acc, RMSE, PFC)
Inferential Hypothesis Testing & Dependence Correction (Two-Tailed Paired t-Tests, Holm Step-Down, Nadeau-Bengio CV Correction)
End